

Roll No.....

ODD MID Term Examination-September (2025)

Btech (Biotech) Ist Semester

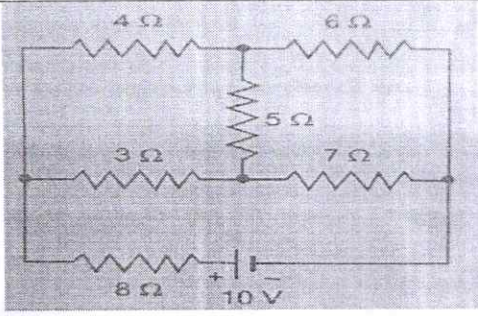
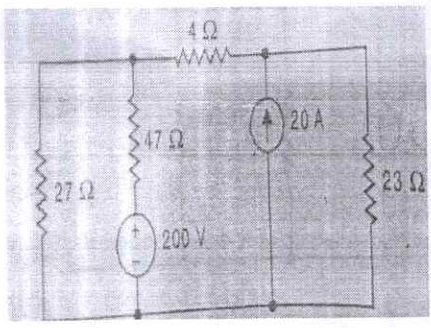
Basic Electrical and Electronics Engineering- EEC-101

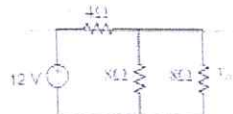
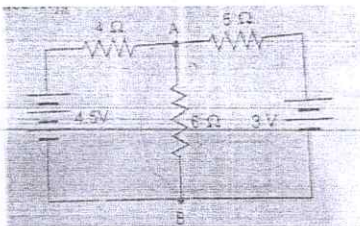
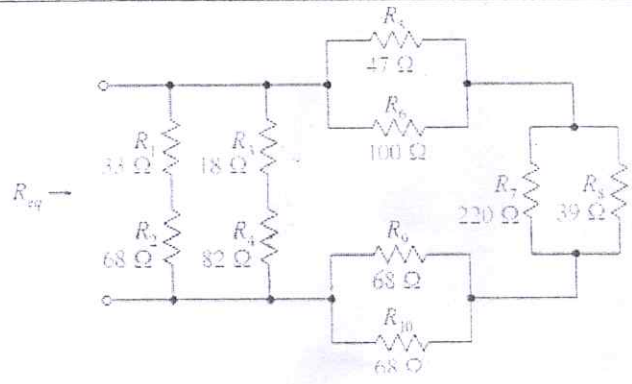
Time: 1.30 Hrs

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing *any one of the sub questions*.
- (ii) Each question carries 10 marks

Q1	(10 marks)	
(a)	 find the current in 5ohm resistance by using KVL or KCL	CO-01
(b)	Define the following terms: i. Active and Passive sources ii. Ohms Law iii. KVL and KCL with an example iv. Ideal and Practical sources v. Unilateral and Bilateral sources	
Q2	(10 marks)	
(a)	Define Superposition theorem find the current in 23 ohm resistance by using Superposition theorem 	CO-01

(b)	For the circuit shown in Figure, use a source transformation to determine v_o 	
Q3	(10 marks)	
(a)	Explain Thevenin's Theorem and find the current in 6 ohm resistance by using Thevenin's theorem 	CO-01
(b)	 find the equivalent resistance of the circuit	
Q4	(10 marks)	CO4
(a)	Explain the working of PN junction diode along with its V-I characteristics.	
(b)	A sample of Ge has intrinsic charge carrier density of $2.0 \times 10^{19}/m^3$. If it is doped with donor impurity of 1 atom in every 10^7 Ge atom. If the Ge atom density is $4.0 \times 10^{28}/m^3$, then determine the majority and minority charge carriers as well as conductivity and resistivity of the doped semiconductor. Given that the mobility of free electrons and holes is $0.35m^2/V\text{-sec}$ and $0.16m^2/V\text{-sec}$ respectively, $n_i = 10^{10}/m^3$.	
Q5	(10 marks)	
(a)	Explain the working of center tapped full wave rectifier and derive the equation for DC current.	CO4
(b)	Explain the formation of Energy band diagram in materials. Classify the materials based on energy band diagram.	